

Telehealth Software Requirement Documentation

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1. Executive Summary

Objective: Build a telehealth system that is easy to use, follows all healthcare rules, and can grow as the clinic expands.

Scope: Covers Patient Intake, Appointment Scheduling, Telemedicine, Clinical Notes, and EHR Integration.

Clinics: 5 locations | 25 doctors | U.S. compliance (HIPAA, HITECH)

2. Market Research Summary

2.1 Market Trends (U.S. Healthcare, 2024-25)

- Telehealth adoption has surged since the COVID-19 pandemic, with **80% of patients open to using virtual care options**.
- The U.S. telehealth market is expected to reach \$90 billion by 2026, driven by convenience, aging populations, and chronic disease management (**source: Statista**).
- 67% of healthcare providers in the U.S. have adopted some form of telemedicine (**source: American Medical Association**).
- Remote monitoring and virtual care tools have shown to reduce hospital readmission rates by 20-25%.
- Increasing focus on value-based care and patient-centered outcomes is accelerating tech adoption.
- Key concerns for providers: maintaining continuity of care, ensuring HIPAA compliance, and integrating with existing Electronic Health Record (EHR) systems.

2.2 Patient Preferences and Digital Health Behavior

- Over **60% of patients prefer online appointment scheduling** over calling a clinic (**source: Accenture**).
- Mobile health app **usage grew by 35% in the past year** alone.
- **Patients aged 18–34 are the most active users** of digital health tools, but telehealth use is growing among seniors too.
- **70% of patients say they want access to their medical records**, lab results, and follow-up instructions via an app or online portal.

2.3 Regulatory and Compliance Considerations

- All telehealth systems must comply with **HIPAA** (Health Insurance Portability and Accountability Act), **ensuring encryption of video sessions and patient data**.
- The **HITECH Act** (Health Information Technology for Economic and Clinical Health Act) **promotes the secure electronic exchange of health information** and penalizes data breaches.

- Some states have rules that say **telehealth appointments must be paid the same as regular**, in person doctor visits.
- **Interstate licensure compacts now allow doctors to treat patients across state lines**, improving scalability.

2.4 Technology Adoption Drivers

- Improved broadband access and **5G rollout support high-quality video streaming**.
- **AI-powered scheduling** and chatbot triage systems improve efficiency.
- Electronic Health Record (EHR) vendors are increasingly offering built-in telehealth modules.
- **Cloud-based solutions reduce setup costs** and allow rapid scaling across multiple clinics.

2.5 Competitor Analysis:

Platform	Key Features	Limitations
Teladoc	Video consults, app-based reminders	Expensive, rigid workflows
Doxy.me	Browser-based, free version	Limited Electronic Health Record integration
SimplePractice	Telehealth + charting	Lacks multi-location customization

3. Functional Modules with Detailed Workflows

3.1 Patient Intake Management

Workflow:



Features:

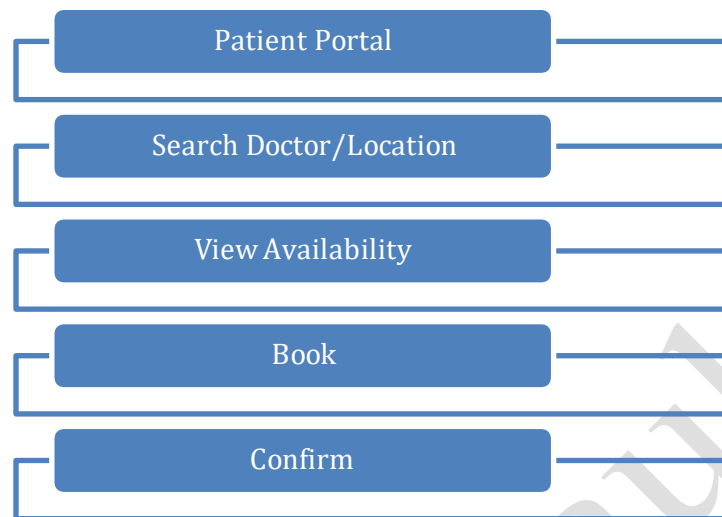
- Digital forms (custom per location)
- Automatically check if a patient's insurance is valid by connecting with insurance companies through a secure system. (**Insurance validation through API**)
- E-signature for consent
- Auto-sync to patient profile

Benefits:

- Reduces manual data entry time
- Error reduction in insurance claims

3.2 Appointment Scheduling

Workflow:



Features:

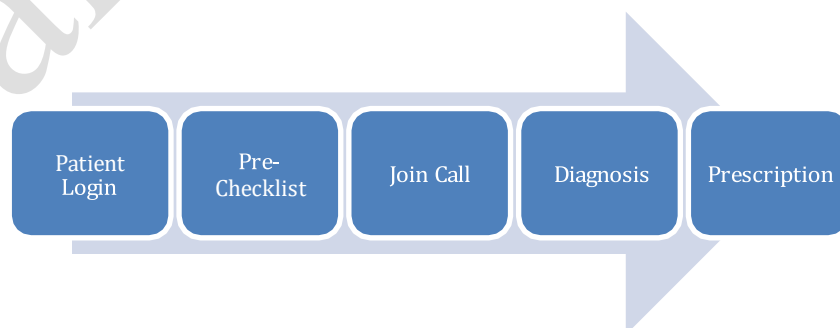
- Real-time slot visibility
- Doctor/location-based filters
- SMS/Email reminders
- Admin privileges to change appointments and reschedule.

Benefits:

- Reduces “no-shows”
- Efficient doctor utilization across 5 locations

3.3 Telemedicine Video Consultation

Workflow:



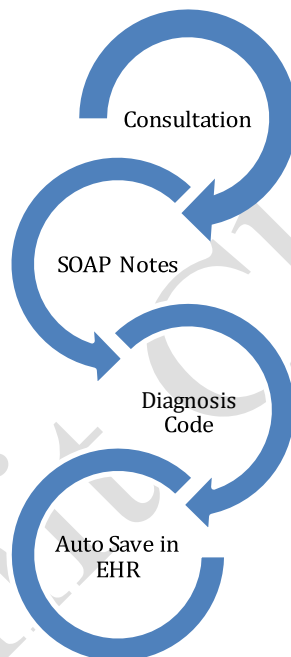
Features:

- Web Real Time Communication-based secure video calls
- In-call notes & attachments
- Prescription generator
- Session recording (with consent of the patient)

Benefits:

- Operational cost saving on physical visits
- Expands reach to rural areas

3.4 Clinical Notes & Charts

Workflow:**Features:**

- Templates for SOAP/PIE/DAR formats
 - SOAP: Subjective, Objective, Assessment, and Plan
(A structured format used by healthcare providers to document patient notes.)
 - PIE: Problem, Intervention, Evaluation
(A method of documenting nursing care in a problem-focused way.)
 - DAR: Data, Action, Response
(A note-taking format used for clinical documentation, especially in nursing.)

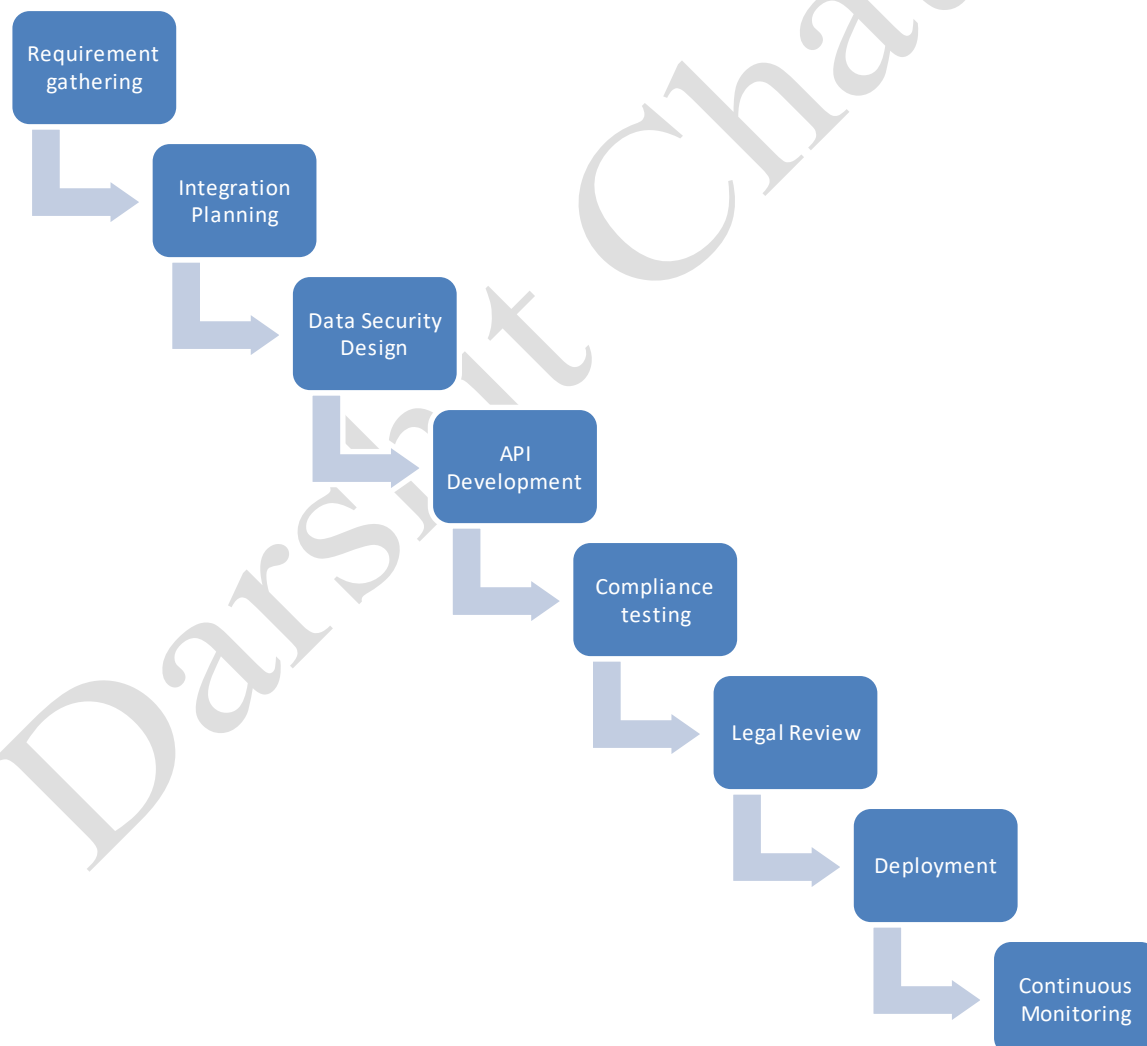
- ICD-10 (International Classification of Diseases, 10th Revision) integration: A coding system used to classify diseases and health conditions for clinical and billing purposes.
- CPT (Current Procedural Terminology) integration: A set of codes used to describe medical, surgical, and diagnostic procedures and services.
- Doctor's Voice-to-text Notes & Medical terms shortform expansion
- Keep a secure record of all changes made in the system, showing who did what and when — with the ability to see past versions if needed. (Trail Auditing)

Benefits:

- Saves consultation time per patient for doctors
- Makes sure the right medical codes are used so the clinic gets paid correctly and avoids billing mistakes.

3.5 Integration and Compliance

Workflow:



Features:

- HIPAA-compliant Software Construction.
- Connection with existing Electronic Health Record (EHR) so that all information stays in one place.
- HL7 Standard: Health Level Seven; A set of international standards for transfer of clinical and administrative data.
- FHIR Standard: Fast Healthcare Interoperability Resources; A standard describing data formats and APIs for exchanging electronic health records.
- Role-based access control (RBAC) to the medical records to ensure patient privacy.

Benefits:

- System follows all laws and regulations to avoid legal issues.
- Easy for doctors to get started and set up at any of the clinic's locations.

4. Suggested Technology Stack

Component	Suggestion	Reason
Frontend	React.js / Flutter	Fast UI, multi-platform
Backend	Node.js / Django	Scalable and secure
Video API	Twilio / Agora	HIPAA-compliant real-time video
Database	PostgreSQL / MongoDB	Structured + flexible storage
Hosting	AWS/GCP (HIPAA-ready)	Scalable, U.S. servers

5. Implementation Roadmap (Phases)

Phase	Milestone	Duration
1	Requirement Gathering & Wireframes (UI Blueprint)	2 weeks
2	Patient Intake + Scheduling	4 weeks
3	Telemedicine & Clinical Notes	5 weeks
4	Integration & Testing	3 weeks
5	Go-live & Feedback Loop	2 weeks

6. KPIs to Track

- Avg. Patient Onboarding Time
- Appointment No-Show Rate
- Consultation Duration
- Doctor Utilization Rate
- Patient Satisfaction Score (Net Promoter Score)

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